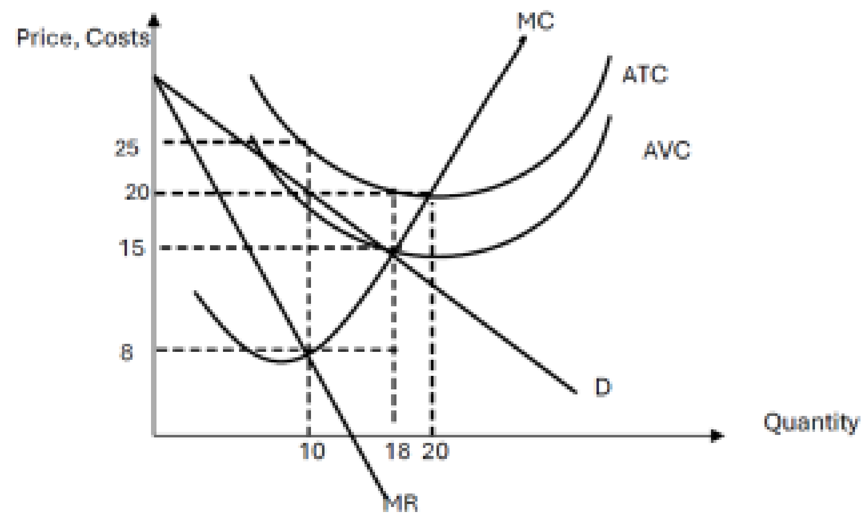
BRAC University
ECO101: Introduction to Microeconomics
Practice Problems Set 2
Spring 2025

Reminders:

- Solve the problems provided in the slides and solved in class for a better understanding of the following problems.
- Do not trust AI fully to solve the problems, as it often gives incorrect answers.

1.



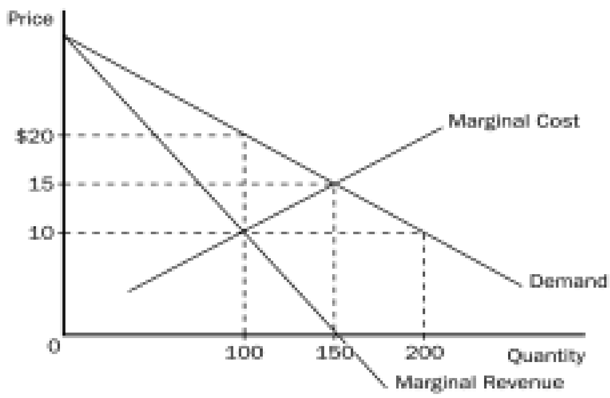
Answer the following question using the graph provided above.

a) What type of competition is the firm facing? b) What is its profit-maximizing level of output and price? c) Is the firm earning a profit or making a loss? Calculate the profit/loss. d) If this firm is operating in a competitive market, what will be the price and output? e) Calculate the amount of loss/profit if this firm is operating in a competitive market. f) How does a monopoly cause a deadweight loss?

2. A cable monopolist has demand, cost and marginal revenue curves given by:

Qd = 1000 - 2P
TC = 5000 + 50Q
MR = 500 - Q

a) Find the cable monopolist's profit-maximizing quantity and price.
b) Find the cable monopolist's resulting profit.



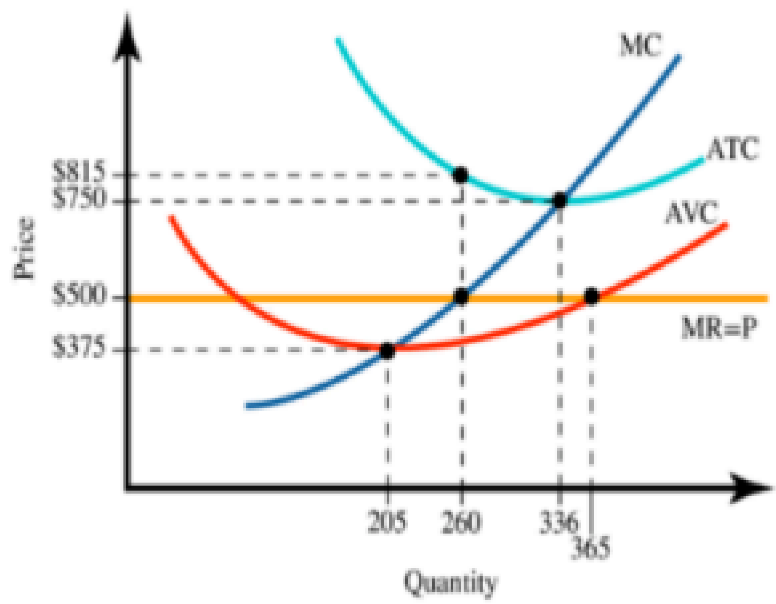
3. Consider the graph above.
a) What will be the output and price if this firm is operating in a competitive market?
b) What will be the output and price if it is a monopoly?
c) What is the amount of deadweight loss in this market if it is a monopoly? d) If the highest price in the market is $30, what will be the consumer surplus if it is a monopoly?

4. In a perfectly competitive industry, all firms have identical cost curves. Suppose that a representative firm's total cost is given by the equation: ����=���� +����+����, where q is the quantity ^{2} of output produced by an individual firm.
Additionally, the market demand for this product is given by the equation **P=1000−2Q** and the market supply is given by ��=������+��, where Q is the overall market output. a) Find the equilibrium price and quantity in the market.

b) The firm's MC equation, based upon its TC equation, is ����=����+��. Given this information and your answer in part (a), what is an individual firm's profit-maximizing level of production in the short run?

c) Calculate the firm's total revenue, total cost, and profit at this profit-maximizing level of production.

d) Find the equation for a representative firm's average total cost (ATC) curve. What should the relationship between price and ATC be in the long run?



5. From the above graph:

a) What is the firm's new profit-maximizing level of output?
b) At this output, calculate the firm’s total revenue, total cost, and profit/loss. c) Is the firm making positive economic profit, zero profit, or a loss in the short run? Should the firm continue to operate in the short run?
d) In the long run, how will the firm adjust?

6. Consider the following table of long-run *total cost* for three different firms:

	Quantity						
	1	2	3	4	5	6	7

Firm A	$60	$70	$80	$90	$100	$110	$120
Firm B	$11	$24	$39	$56	$75	$96	$119

For each of these firms, mention whether they follow economies, diseconomies, or constant returns to scale. Be sure to show your calculations for the first three levels of output.

7. Bob produces DVD movies for sale, which require only a building and a machine that copies the original movies into a DVD. Bob rents a building for $30,000 per month and rents a machine for $20,000 a month. These are his fixed costs. His variable costs are given in the following table:

Quantity of DVDs	**VC ($)**
0	0
1,000	5,000
2,000	8,000
3,000	9,000
4,000	14,000
5,000	20,000
6,000	33,000
7,000	49,000
8,000	72,000
9,000	99,000
10,000	1,50,000

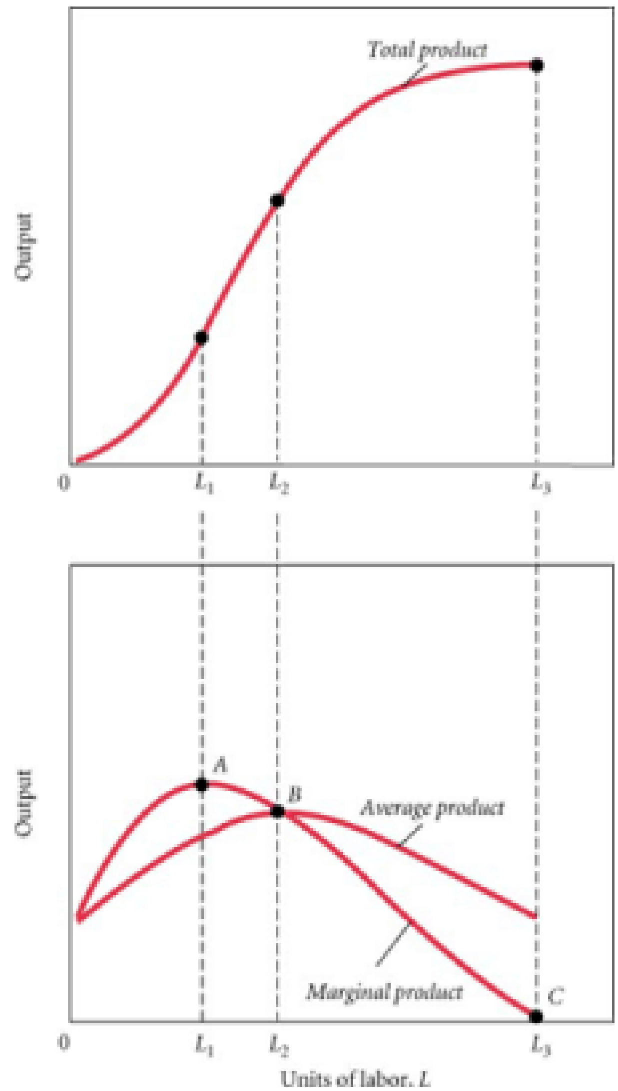
a) Calculate *TC*, *AVC*, *ATC,* and *MC* for each quantity of output. Copy and add the table columns accordingly to show your calculated values. You do not need to show your working for all the values.

b) Identify Bob's break-even price (zero economic profit) and shut-down price here. Roughly show the two prices in two separate graphs.

c) Suppose that the current price of a DVD is $23. What is the profit-maximizing quantity of DVDs that Bob should produce? What will be Bob's profit? In the short run, should he produce or shut down? What happens in the long run?

d) Suppose the price of a DVD becomes $20. What is the profit-maximizing quantity of DVDs that Bob should produce? What will be Bob's profit now?

8. Consider the following production function:



a) Explain why the marginal product curve reaches its peak at point A.

b) What is the algebraic representation for calculating marginal product? c) Draw a generic total cost curve that is consistent with the production function depicted above.

d) After writing the algebraic representation of the slope of a total cost curve, explain what it tells you.

e) Redraw the production function for an improvement in technology.

9. The "Greasy Wrench" shop, which specializes in auto repairs, rents a new building. In the short run, it can vary its output (the number of cars fixed) only by varying the number of

mechanics it employs. Each Greasy Wrench employee is paid a fixed wage. Over the years, Wu Wei, the owner of "Greasy Wrench," has tried hiring different numbers of workers.

L	AP	TP	MP
1	5		
2	7		
3	10		
4	11		
5	10		
6	9		

a) Complete the table by calculating the Total Product and Marginal Product corresponding to each number of laborers.

b) At which worker does the Law of Diminishing Marginal Returns start working? Explain your reasoning.

10. An industry currently has 500 firms, each of which has fixed costs of Tk. 20 and marginal costs as follows:

Quantity	Marginal cost (in Tk.)
1	20
2	15
3	12
4	15
5	20
6	26

a) Figure out the firm's total cost, variable cost, average variable cost, and average total cost for each quantity.

b) The equilibrium price is currently Tk. 12. Given it's a perfectly competitive market, how much does each firm produce? A diagram must be illustrated.

c) At what price will the firm make a normal profit? At what output will the firm decide to close down its production? Show the answer in a diagram.